faster than nearby ones. So, by measuring a galaxy's speed of recession (another simple task), one can deduce a galaxy's distance.

If those two well-tested methods give different distances for the same object, something must be wrong. Either the supernovas are bad standard candles, or our model for the rate of cosmic expansion as measured by galaxy speeds is wrong.

Well, something *was* wrong. It turned out that the supernovas were splendid standard candles, surviving the careful scrutiny of many skeptical investigators, and so astrophysicists were left with a universe that had expanded faster than we thought, placing galaxies farther away than their recession speed would have otherwise indicated. And there was no easy way to explain the extra expansion without invoking lambda, Einstein's cosmological constant.

Here was the first direct evidence that a repulsive force permeated the universe, opposing gravity, which is how and why the cosmological constant rose from the dead. Lambda suddenly acquired a physical reality that needed a name, and so "dark energy" took center stage in the cosmic drama, suitably capturing both the mystery and our associated ignorance of its cause. Perlmutter, Schmidt, and Reiss justifiably shared the 2011 Nobel Prize in physics for this discovery.

The most accurate measurements to date reveal dark energy as the most prominent thing in town, currently responsible for 68 percent of all the mass-energy in the universe; dark matter comprises 27 percent, with regular matter comprising a mere 5 percent.

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The shape of our four-dimensional universe comes from the relationship between the amount of matter and energy that lives in the cosmos and the rate at which the cosmos is expanding. A convenient mathematical measure of this is omega: Ω, yet another capital Greek letter with a firm grip on the cosmos.

If you take the matter-energy density of the universe and divide it by the matter-energy density required to just barely halt the expansion (known as the "critical" density), you get omega.

Since both mass and energy cause space-time to warp, or curve, omega tells us the shape of the cosmos. If omega is less than one, the actual mass-energy falls below the critical value, and the universe expands forever in every direction for all of time, taking on the shape of a saddle, in which initially parallel lines diverge. If omega equals one,